

DreemFolio AI — Enterprise Platform Manual

Comprehensive User Guide, Authentication Workflows, System Validations & 100% Feature Suite

Version: 2.3.0 • **Target Audience:** New Candidates, Experienced Job Hunters & System Administrators

System Module	Target Capabilities	Key Output
Authentication & Security	Bcrypt, JWT Tokens, Google 1-Tap OAuth, Sliding Rate Limits	Authenticated Profile & Secure Session
AI Resume Tailor	Gemini Flash + Google XYZ Formula + Archetype Engine	Role-Optimized ATS JSON & Metrics
Dual PDF Engine	ReportLab Flowables & Responsive Canvas Renderer	100% Machine-Readable ATS & Visual PDFs
Cover Letter Studio	Tone-Adaptive Generative Engine + Digital Signatures	Tailored Employer Letter PDF
Portfolio Studio	Bento Grid & CSS Theme Generator + Custom Domains	Live Cloud Website (`/p/slug`) & ZIP
Security & QR Card	SVG/PNG QR + Digital vCard (.vcf) + 4-Digit Encrypted PIN	Instant Mobile Contact & Anti-Bruteforce
Job Hunter & Copilot	Live Market Search & Kanban Status Tracker	Tailored Outreach Kits & Status Board
Mock Voice Studio	Web Audio Speech + STAR Evaluator + Speech Pace WPM	Evaluation Scorecard & PDF Report
AI Video Conference	Gemini Live Multimodal 24kHz WebSocket (<800ms)	Oral Video Interview + Real-Time Mistake HUD
Admin Panel	FastAPI RBAC + Country Localization + Quota Overrides	User Management & Global Price Matrix

■ **Executive Summary for New Users:** DreemFolio AI is engineered as an enterprise-grade career accelerator. New users can register in seconds with Email or Google, upload their existing resume, paste a target Job Description, and receive a 100% ATS-compliant resume and cover letter formatted with the Google XYZ formula. Users can publish a live developer portfolio with QR smart card integration, and practice live with our real-time Gemini Live AI video interviewer with instant mistake coaching before facing hiring managers.

1. Account Setup, Registration, Login & Strict Validations

1.1 Registration Workflow & Field Validations

New users can register by clicking 'Sign In / Register' in the top header and choosing the Register tab:

- **Full Name:** String, maximum 100 characters. Defaults to 'Candidate' if left blank.
- **Email Address:** Must be a valid email format ('user@domain.com'). Automatically lowercased and stripped of spaces. Duplicate emails return HTTP 400 ('An account with this email address already exists').
- **Password Security:** Strict minimum length of 6 characters required. Encrypted using industry-standard `bcrypt` hashing with salt rounds.
- **Automated Login:** Upon successful registration, the system creates the user on the Free Starter tier, generates a cryptographic JWT access token, and logs the user in without requiring manual re-entry.

1.2 Sign-In / Login Workflow & Security Protocols

- **Authentication Check:** Compares entered password against stored bcrypt hash. Invalid credentials return HTTP 401 ('Invalid email or password').
- **Account Status:** Deactivated accounts return HTTP 403 ('Your account has been deactivated').
- **JWT Session Storage:** Access token is stored in browser `localStorage` as `saas\_auth\_token` and passed as `Authorization: Bearer `.
- **Google OAuth2 / 1-Tap Sign-In:** Direct native popup verifies the Google ID token server-side. Auto-creates account with Google avatar and full name.
- **Anti-Bruteforce Protection:** Strict sliding-window rate limiters restrict login to 10 attempts/min and registration to 5 attempts/min per IP (HTTP 429).

2. AI Resume Builder & 100% ATS Optimizer

2.1 Upload & High-Precision Parsing

- **Formats Supported:** PDF, DOCX, and TXT files up to 10MB limit (HTTP 413 if exceeded).
- **Magic-Bytes Verification:** PDF files are verified against `%PDF-` binary header to prevent malicious disguised uploads.
- **Extraction Engine:** Automatically extracts Candidate Name, Email, Phone, Location, Summary, Experience, Education, Projects, and Skills.

2.2 Target Job Alignment & The Google XYZ Formula

When pasting a target Job Description (JD), the platform performs deep semantic keyword analysis:

- **Keyword Matching:** Highlights matching skills and pinpoints critical missing technical competencies.
- **Google XYZ Bullet Rewriting:** Transforms passive task descriptions into quantified executive statements: *'Accomplished [X], as measured by [Y], by doing [Z]'*.
- **Role Archetypes:** Supports 6 specialized archetypes (Software Engineering, QA & Automation, Automotive Technician, Data Science/AI, Product Management, and Executive).

2.3 Live Real-Time ATS Audit Checklist

The right panel displays a live ATS score (0-100), formatting checklist, action verb count, and quantified impact metrics score.

3. Dual PDF Generation Engine (ATS vs Visual)

Engine A: Single-Column ATS PDF (Recommended for Online Portals)

Built with Python ReportLab flowables following strict machine-readability guidelines. It uses standard font metrics, sequential vertical reading flow, and zero complex multi-column floating boxes, ensuring 100% parsing success across Workday, Taleo, Greenhouse, and Lever.

Engine B: Modern Visual Executive PDF (Recommended for In-Person & Executive Pitches)

Engineered for human hiring managers and portfolio presentations. Features subtle colored section accents, modern typography, elegant margins, and high-impact visual hierarchy (Indigo Banner, Visual Sidebar, Emerald Prestige, Tech Noir).

Interactive Preview Features:

- **Dual-Mode Toggle:** Switch between PDF.js Canvas HD vector rendering and pure fallback Image rendering.
- **Zoom & Page Controls:** Zoom in/out from 50% to 200% with multi-page counter navigation.

4. Interactive Portfolio Studio, QR Smart Card & PIN Security

4.1 Instant Portfolio Generation & Themes

With 1 click, your tailored resume is converted into a fully responsive developer website with themes: Bento Grid, Cyberpunk, Minimalist Clean, and Classic Executive.

4.2 Cloud Publishing & Custom Domains

- **Instant URL:** Published to <https://dreemfolio.com/p/{slug}> with automatic global CDN caching.
- **Custom Domain:** Connect your own domain (e.g. portfolio.yourname.com) via simple CNAME mapping.

4.3 QR Smart Card, Digital vCard & PIN Security

- **Dynamic QR Code:** Generate crisp SVG and PNG QR codes linking directly to your live portfolio.
- **Digital vCard (.vcf):** Allows recruiters to save your phone, email, and social links to their phonebook with 1 tap.
- **4-Digit PIN Protection:** Protect confidential portfolio projects and contact details with a 4-digit PIN, backed by anti-bruteforce lockouts (10 attempts/min).

5. AI Job Hunter, Application Copilot & Kanban Tracker

5.1 Live Job Search Engine: Search global remote and local positions filtered by Work Mode and Experience Level.

5.2 1-Click Application Copilot Kit: Generates Tailored Cold Emails, LinkedIn InMail scripts, 30-second Elevator Pitches, and Salary Negotiation scripts.

5.3 Job Application Kanban Tracker: Move applications across columns: Saved -> Applied -> Interviewing -> Offered -> Archived.

6. AI Voice Mock Interview & Real-Time Video Conference

6.1 AI Voice Mock Interview Studio

- **Oral Question Delivery:** The AI interviewer speaks questions aloud through browser speech synthesis.
- **Microphone Answer Transcription:** Speaks your answer naturally; real-time SpeechRecognition transcribes your speech.
- **STAR Framework Scoring:** Grades each answer on Situation, Task, Action, and Result (1–10 scale).
- **Speech Analytics:** Measures Words Per Minute (WPM) speaking pace and flags filler words (um, uh, like).

6.2 Real-Time Video Conference Studio (Alex / Zoom Style)

The platform's flagship interactive stage simulating a real video call with an AI lead interviewer:

- **Google Gemini Live WebSocket Engine:** Powered by gemini-2.5-flash-native-audio-latest over bidirectional WebSockets, providing instant voice-to-voice speech with sub-second latency (< 800ms).
- **Studio Audio Quality:** Streams 24,000Hz 16-bit linear PCM audio with zero intermediary encoding delays.
- **Voice Selector:** Switch between Alex (Puck Male), Aoede (Female), Charon (Deep Male), Kore (Natural Female), and Fenrir (Executive Male).
- **Live Mistake Coaching (Waradi Kiyala Denna):** Floating HUD alerts drop down instantly if you make a technical error or omit safety protocols.
- **Mobile Audio Unlock:** Single tap on '■ Tap to Hear / Replay' unlocks the Web Audio API on iOS Safari and Android Chrome.

7. Super Admin Control Panel, Validations & System Rate Limits

7.1 Admin Console (/paneladmin)

Accessible to administrators for full platform oversight: Manage registered accounts, 1-Click Plan Overrides (Free, Pro, Sprint Pass, Lifetime), and Global Country Pricing matrix (adjust purchasing power parity for Sri Lanka, India, etc.).

7.2 System Error Codes & Validations Summary Table

Action	Condition / Rule	HTTP Code & Response
Registration	Duplicate email	400: Email already exists
Registration	Password < 6 chars	422: Minimum 6 characters required
Login	Wrong password / email	401: Invalid credentials
Resume Upload	File > 10MB	413: Size exceeds 10MB limit
Resume Upload	Corrupted PDF header	400: Invalid PDF format (%PDF-)
AI Tailor	Empty JD or Resume	400: Text cannot be empty
AI Tailor	Exceeded daily quota	402: Daily quota exhausted
PIN Verify	> 10 attempts/min	429: Too many requests

End of Operational Manual — DreemFolio AI Enterprise Platform (v2.3.0)